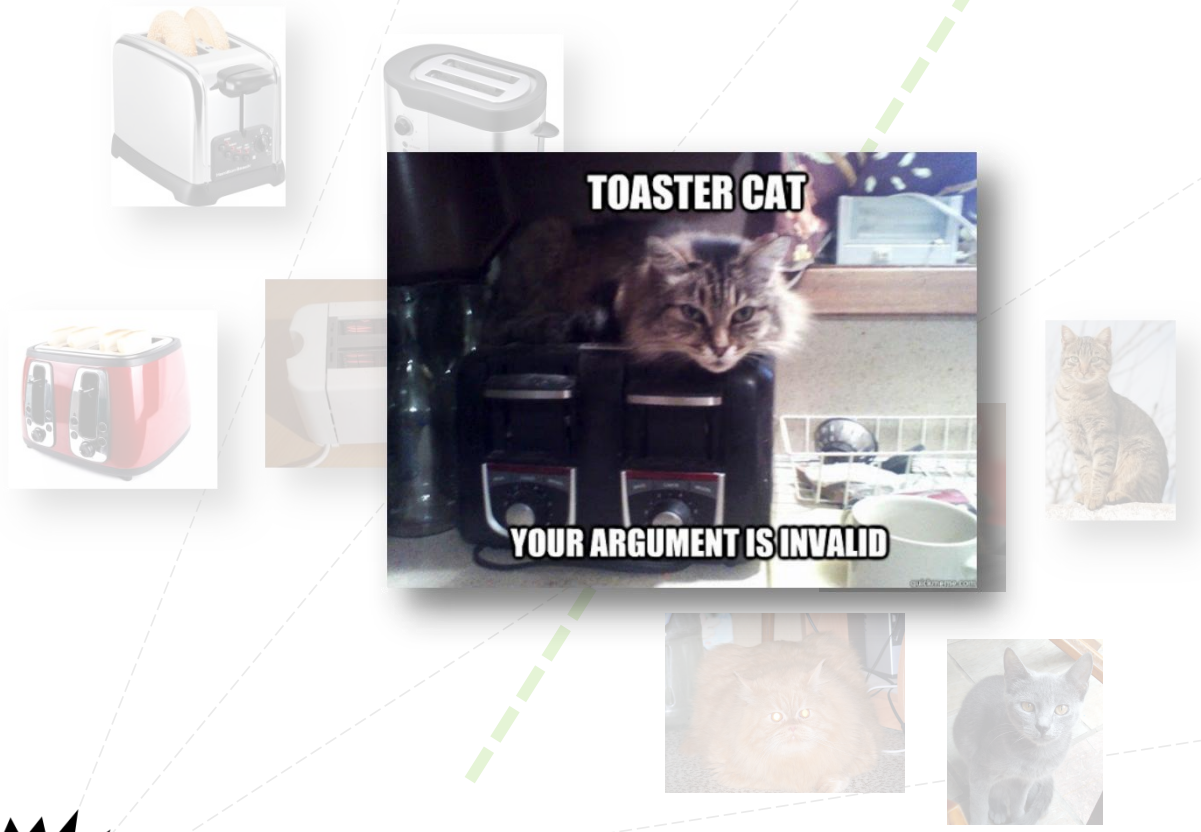
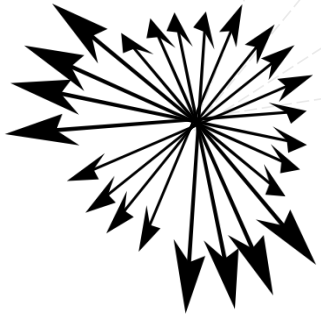
Images As High-Dimensional Vectors

- An image is just a bunch of numbers
- Let's stack them up into a vector
  - Our training data is just a bunch of high-dimensional points now

- Divide space into different regions for different classes





The Space of All Images

Images As High-Dimensional Vectors

- An image is just a bunch of numbers
- Let's stack them up into a vector
  - Our training data is just a bunch of high-dimensional points now

- Divide space into different regions for different classes

or

- Define a distribution over space for each class

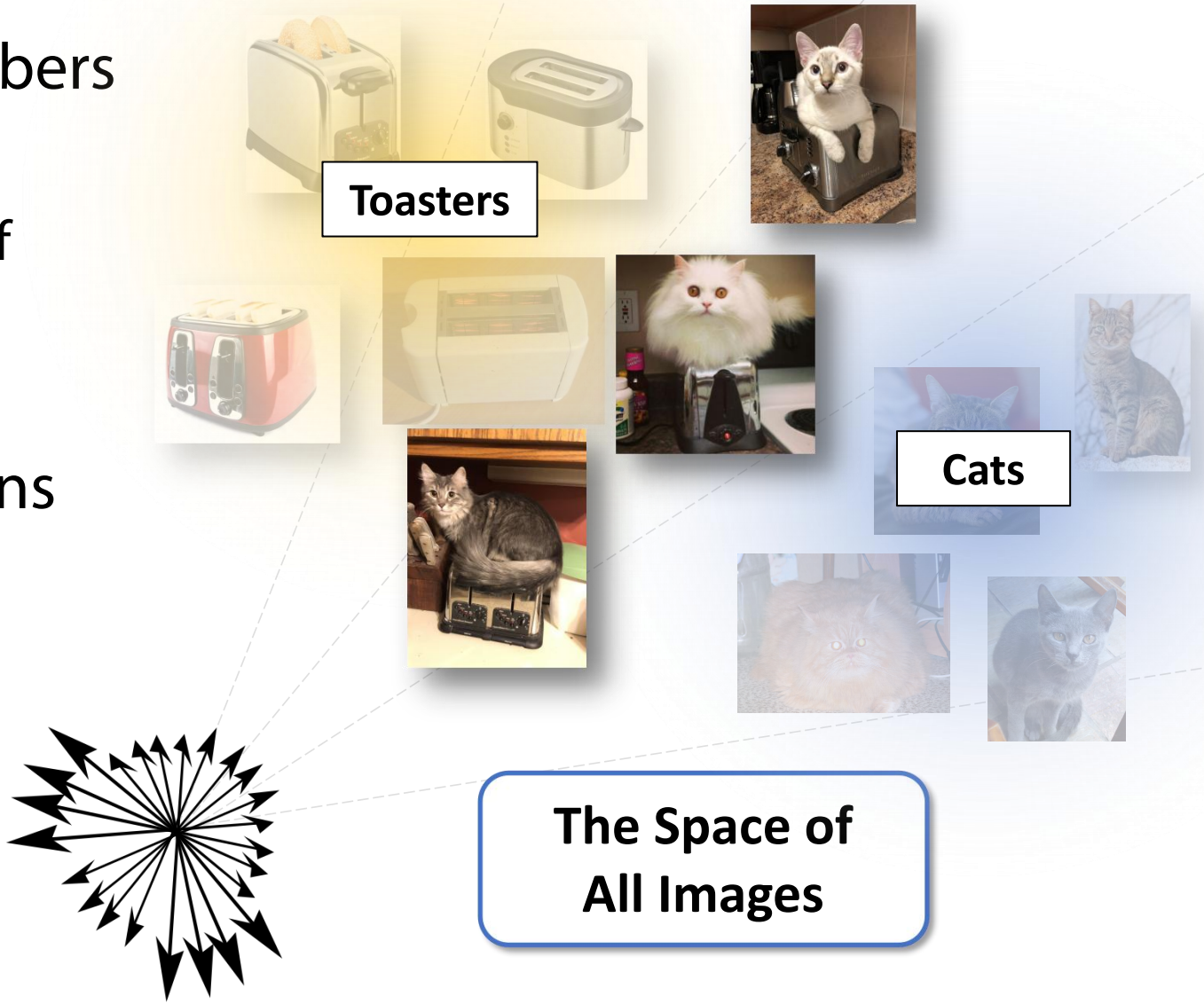


Image Features and Dimensionality Reduction

- How high-dimensional is an image?
  - Let's consider an iPhone X photo:
    - 4032 x 3024 pixels
    - Every pixel has 3 colors
    - 36,578,304 pixels (36.5 Mega pixels)
- In practice, images sit on a lower-dimensional manifold
- Think of image features and dimensionality reduction as ways to represent images by their location on such manifolds

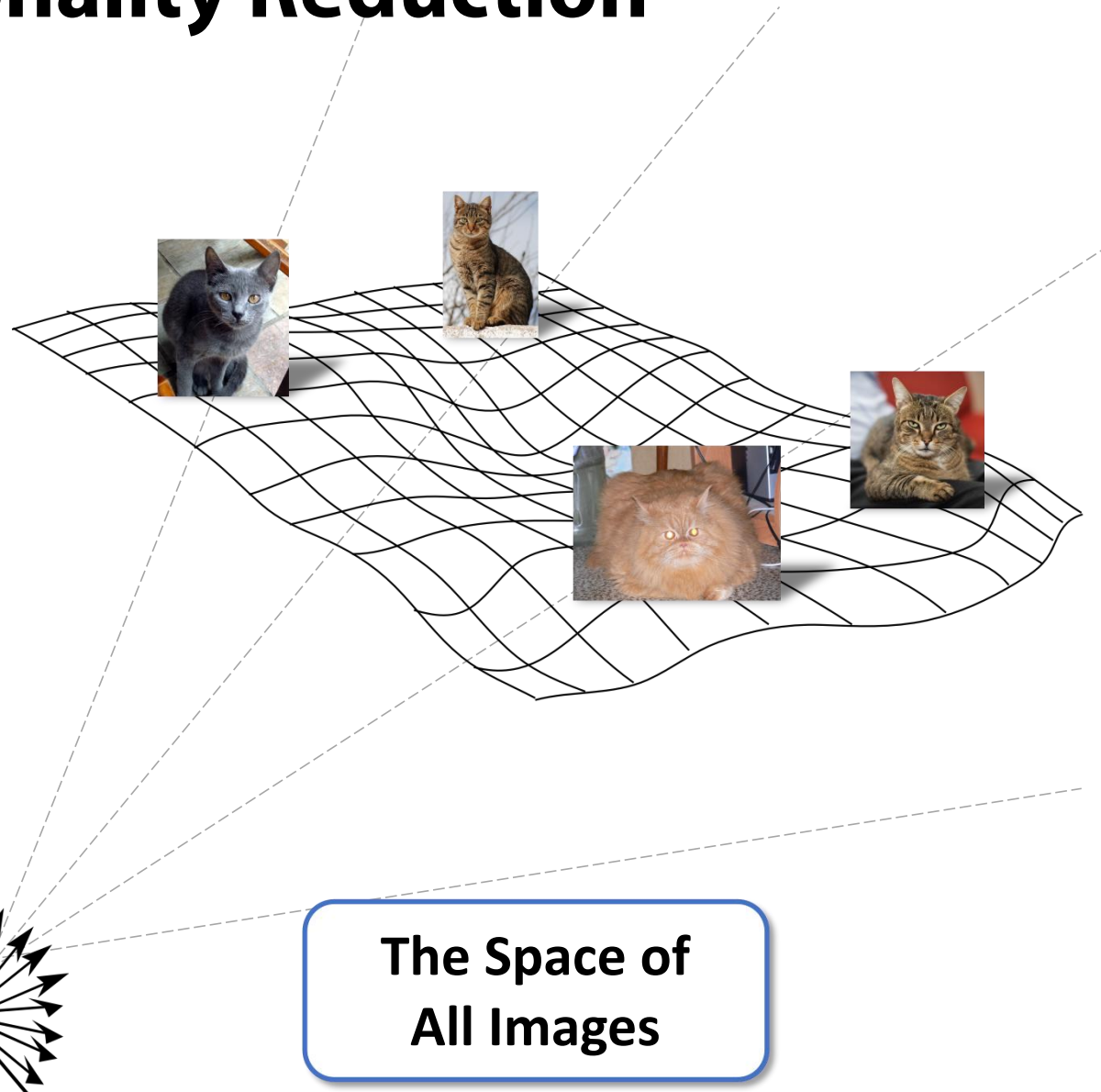
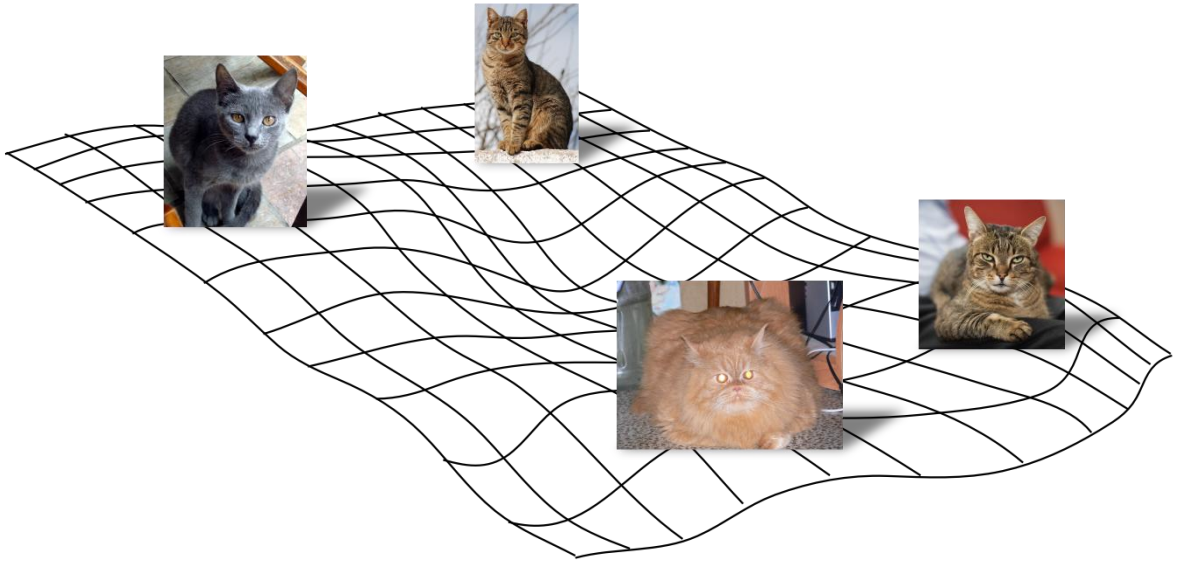


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Side Note:
This also lets us deal with images of different sizes, crops, etc.

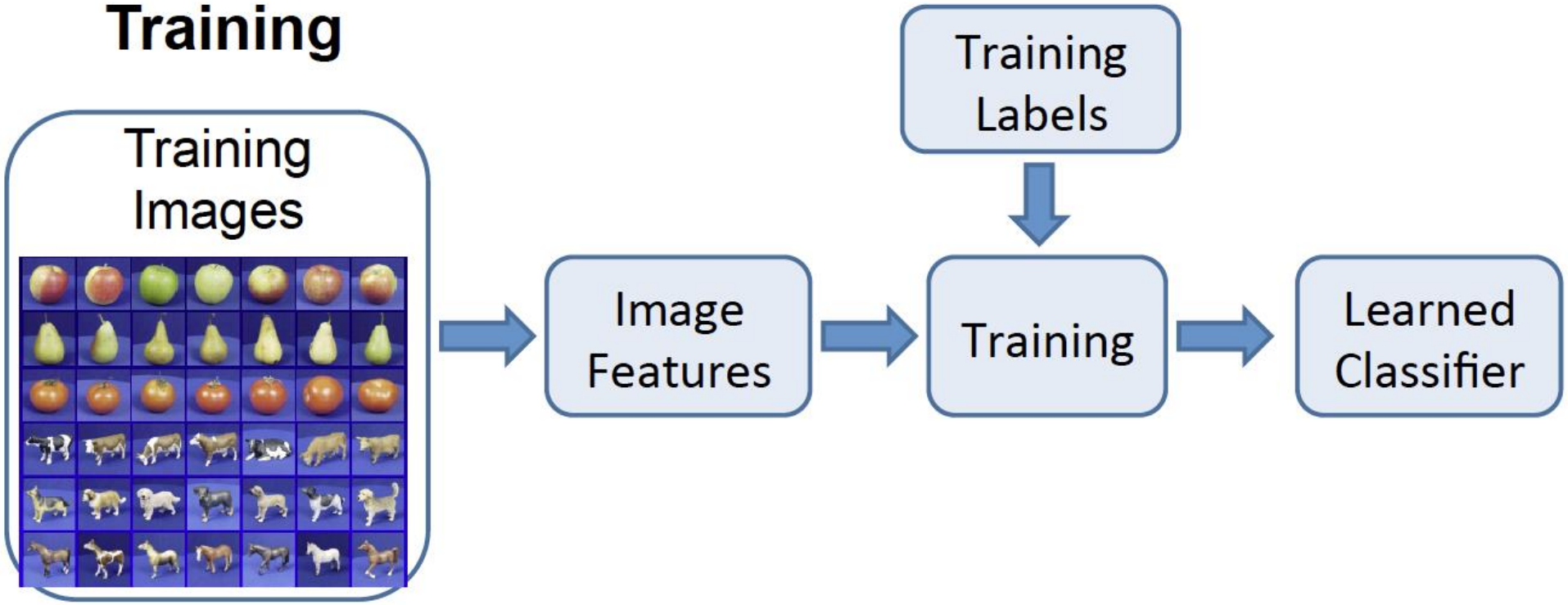
Training & Testing a Classifier

- Collect a database of images with labels
- Use ML to train an image classifier
- Evaluate the classifier on test images



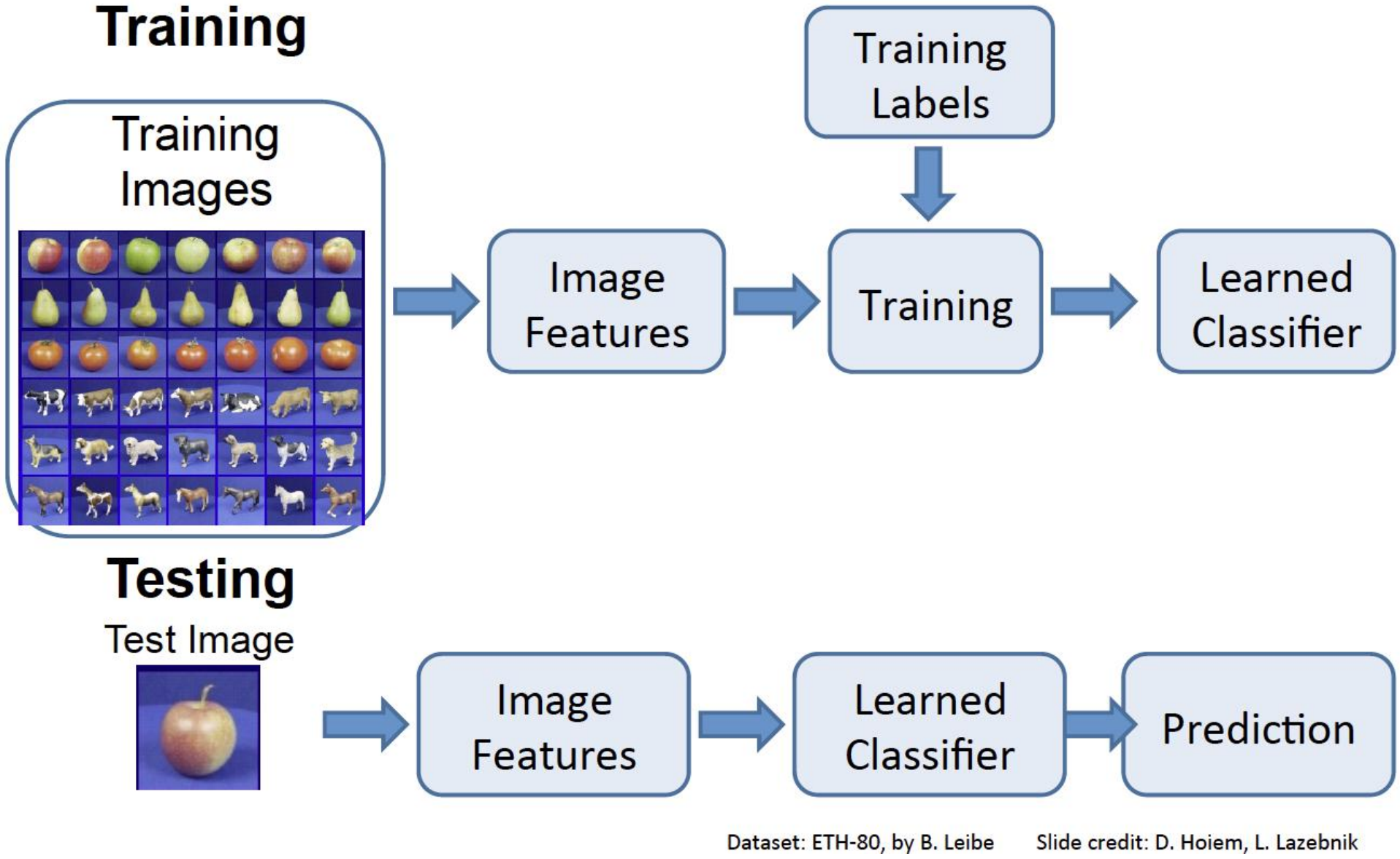
Slide from Andrej Karpathy and Fei-Fei Li http://vision.stanford.edu/teaching/cs231n/

Training & Testing a Classifier



Dataset: ETH-80, by B. Leibe Slide credit: D. Hoiem, L. Lazebnik

Training & Testing a Classifier



Dataset: ETH-80, by B. Leibe Slide credit: D. Hoiem, L. Lazebnik

Classifiers

- Nearest Neighbor
- kNN (“k-Nearest Neighbors”)
- Linear Classifier
- Neural Network
- Deep Neural Network
- Transformers
- …